SMARTBRIDGE
Let's Bridge the Gap

Date	28-10-2026
Team ID	**EN2EL3010078**
Project Name	EV Market Performance and Insights
Maximum Marks	4 Marks

Project Design Phase - I

Solution Architecture

Solution Architecture:
Solution architecture is a complex process - with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

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Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	The EV market dataset that holds vehicle sales, price, battery, range, charging, manufacturer, country, rating and CO2 information.	**Excel / CSV file**
Data Connection	Connects Tableau to the dataset and lets the data be refreshed when the source file changes.	**Tableau Desktop data source connection**
Data Preparation	Cleans the data, sets data types, relates tables and assigns geographic roles for country and region fields.	**Tableau Data Source tab**
Data Layer	Stores the prepared data model and extract that every worksheet and dashboard reads from.	**Tableau data model / extract**
Analytics / BI Layer	Calculates KPIs and measures such as total sales, average price, average range, CO2 saved, retained value and efficiency, and applies filters and aggregations.	**Tableau calculated fields and LOD expressions**

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Visualization / Reporting	Presents the insights as charts in two interactive dashboards with a shared category legend, filters and tooltips.	**Tableau Desktop (Dashboard 1 and Dashboard 2)**
Business Users	Strategy directors, BI managers and researchers explore the dashboards to compare segments, manufacturers and technologies and to make decisions.	**Tableau dashboards (Tableau Public / browser)**